

Lie Groups and Lie Algebras

Grant Talbert

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MA731

Abstract

MA731: Lie Groups and Lie Algebras was taught by [Siu-Cheong Lau](#) in the spring 2026 semester (my junior year). We followed [\[Hal15\]](#) in this course. At times, Professor Lau defined interesting topics that are generally not needed for this course, so these notes are laced with interesting, let not always useful, definitions of more advanced concepts.

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Chapter 1

General Theory

1.1 Matrix Lie Groups

Lecture 1: Syllabus day

Tue 20 Jan 2026 10:59

Tuesday 1/27 we have class in cds548 for a seminar on derived symplectic structure.

Definition 1.1.1. A Lie Group G is a group object in the category Man of smooth manifolds. In particular, it is a group G with a smooth structure that is compatible with the group structure in the sense that multiplication $\mu : G \times G \rightarrow G$ is smooth.

Consider for example the group $G = \text{SU}(2)$ of Hermitian matrices $AA^\dagger = 1$. We have $\text{SU}(2) \cong S^3$ as spaces, so there exists a manifold structure on $\text{SU}(2)$. It then becomes a Lie group.

Example 1.1.2 (Matrix groups). Consider $\text{GL}(n, \mathbb{R}) := \text{Aut}_{\mathbb{R}\text{-Vect}}(\mathbb{R}^n)$. This is an open subset of $\mathbb{R}^{n^2}$, and thus has an n^2 -manifold structure. We can also consider $\text{GL}(n, \mathbb{C})$ and obtain an open subset of $\mathbb{C}^{n^2}$. This is a subgroup of $\text{GL}(2n, \mathbb{R})$ in the obvious way. Note that $\text{GL}(n, \mathbb{C})$ acts on $\mathbb{C}^n$ by multiplication, and thus acts on $\mathbb{R}^{2n}$ via the identification $\vec{x} + i\vec{y} \leftrightarrow (\vec{x}, \vec{y})$.

More explicitly, take $\mathcal{A} \in \text{GL}(n, \mathbb{C})$ and $z \in \mathbb{C}^n$. Then we can write $\mathcal{A} = A + iB$ and $z = x + iy$. Then

$$\mathcal{A}z = (Ax + iAy) + (-By + iBx).$$

This can be rewritten of the form

$$\begin{pmatrix} A & -B \\ B & A \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

This is equivalent to the statement that the matrix commutes with the matrix

$$J = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}, \quad I \text{ the } 2n\text{-identity matrix.}$$

We say it *commutes with the complex structure* of multiplication by i .

Continuing to work over $\mathbb{C}^n$, a function is holomorphic when $\partial_{\bar{z}}f = 0$, meaning $\frac{1}{2}(\partial_x + i\partial_y)f = 0$. The reason there is a plus for $\bar{z}$ is because this is dual to $dz = dx + idy$ and $d\bar{z} = dx - idy$ or something.

Using the notion of holomorphy, we can define complex manifolds.

Definition 1.1.3. A complex n -manifold is a topological manifold M with charts $M \supseteq U \cong V \subseteq \mathbb{C}^n$ such that the coordinate transformations are holomorphic.

Holomorphic functions are automatically smooth. Also note that the transition functions are holomorphic *if and only if* $d(\text{transition}) \in \text{GL}(n, \mathbb{C})$.

Consider $\text{SL}(n, \mathbb{R}) \subseteq \text{GL}(n, \mathbb{R})$ of matrices with determinant 1. In particular,

$$A^* (dx^1 \wedge \cdots \wedge dx^n) = \det A (dx^1 \wedge \cdots \wedge dx^n) = dx^1 \wedge \cdots \wedge dx^n.$$

Thus A preserves the volume form.

We can also consider $\text{O}(n, \mathbb{R}) \subseteq \text{GL}(n, \mathbb{R})$ defined by $A^t A = 1$. We also have $\text{SO}(n) = \text{SL}(n, \mathbb{R}) \cap \text{O}(n, \mathbb{R})$. We also have $\text{U}(n, \mathbb{C})$ defined as the set of matrices with $UU^\dagger = 1$, and $\text{SU}(n) = \text{SL}(n, \mathbb{C}) \cap \text{U}(n, \mathbb{C})$.

It is at this point relevant to recall the Riemannian metric $g(x, y) = x \cdot y = x^t y$ on $\mathbb{R}^n$ and the Riemannian metric $g(x, y) = x^\dagger y$ on $\mathbb{C}^n$. We usually write, for $v, w \in \mathbb{C}^n$, with $v = a + bi$ and $w = c + di$,

$$v^\dagger w = g\left(\begin{pmatrix} a \\ b \end{pmatrix}, \begin{pmatrix} c \\ d \end{pmatrix}\right) + \omega\left(\begin{pmatrix} a \\ b \end{pmatrix}, \begin{pmatrix} c \\ d \end{pmatrix}\right)$$

where $\omega(v, w)$ is defined by

$$\omega(v, w) = \begin{pmatrix} a^t & b^t \end{pmatrix} \begin{pmatrix} 0 & -I \\ -I & 0 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix}.$$

We call ω a *symplectic form*.

These structures are related by $\omega(v, w) = g(Jv, w)$, and thus if a map preserves the metric and the (complex) manifold structure, it preserves the Hermitian inner product and symplectic structure.

Definition 1.1.4 ([Huy04]). Let V be a vector space. A *almost complex structure* on V is a map $J : V \rightarrow V$ with $J^2 = -1$. An almost complex structure on an inner product space is *compatible* with the inner product if $\langle J(-), J(-) \rangle = \langle -, - \rangle$. An *almost complex structure* on a manifold X is a vector field $J \in \mathfrak{X}(X)$ such that for all $x \in X$, the map J_x is an almost complex structure on $T_x X$.

Definition 1.1.5 ([Huy04]). Let X be a complex manifold. A Riemannian metric g on X is a *hermitian structure* on X if for any $x \in X$, the inner product $g_x : T_x X \otimes T_x^* X \rightarrow \mathbb{R}$ is compatible with the complex structure J_x . The $(1, 1)$ -form $\omega := g(J(-), -)$ is then called the *fundamental form*.

Let g be the canonical Riemannian metric on $\mathbb{R}^n$. On $\mathbb{C}^n$, we can write the Riemannian metric $h(x, y) = x^\dagger y$ as follows. let $x = a + bi$ and $y = c + di$. We can interpret $x = (a, b) \in \mathbb{R}^{2n}$ and similarly for y in order to define $g(x, y)$. We can thus write

$$h(x, y) = g(x, y) + \omega(x, y),$$

$$\omega(x, y) = \begin{pmatrix} a^t & b^t \end{pmatrix} \begin{pmatrix} 0 & -I \\ -I & 0 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a^t & b^t \end{pmatrix} \begin{pmatrix} -d \\ -c \end{pmatrix} = -a^t d - b^t c.$$

Note that ω is a symplectic form.

Definition 1.1.6 ([Lee00] 22). Let M be a manifold. A *symplectic form* ω on M is a 2-form such that for all $p \in M$, the vector ω_p is a nondegenerate 2-covector, meaning for all $v \in T_p M$ there exists $w \in T_p M$ such that $\omega_p(v, w) \neq 0$.

Note that $\omega(x, y) = g(J(x), y)$. Thus a map preserves h and the complex manifold structure if and only if it preserves the Hermitian inner product and symplectic structure.

1.2 Lie algebras

Lecture 2: The Lie algebra and the exponential

Thu 22 Jan 2026 10:57

Let G a Lie group. Then since left- and right-multiplication $\mathcal{L}_g : h \mapsto gh$ and $\mathcal{R}_g : h \mapsto hg$ are diffeomorphisms, we obtain $T_g G \cong T_e G$ for all $g \in G$ by passing to the pushforwards. Thus there exists a global tangent frame by a choice of basis for $T_e G$. We then identify $X \in T_e G$ with *left-invariant vector field*. This is a vector field which is invariant under left-multiplication, meaning

$$X_g := (\mathcal{L}_g)_* X.$$

As explored in MA721 (see [Lee00] 8.33 and onwards), the set of left-invariant vector fields forms a Lie algebra. Hence we define

Definition 1.2.1. Let G be a Lie group. The *Lie algebra* of G , often denoted $\mathfrak{g}$, is defined to be the set of left-invariant vector fields of G , and is generally identified with $T_e G$.

Definition 1.2.2. Let X be a manifold with hermitian structure. If the fundamental form ω is closed, i.e. $d\omega = 0$, then the manifold is called a *Kähler manifold* and ω is called the *Kähler form* ([Huy04] 3.1.6). A *Calabi-Yau manifold* is a compact Kähler manifold with a nowhere-vanishing holomorphic n -form.

Definition 1.2.3. A *Lie groupoid* is a groupoid $\mathcal{G}$ such that $\text{Obj}(\mathcal{G}), \text{Mor}(\mathcal{G})$ are smooth manifolds, the maps $\text{dom}, \text{cod} : \text{Mor}(\mathcal{G}) \rightarrow \text{Obj}(\mathcal{G})$ are smooth, identity $1_\bullet : \text{Obj}(\mathcal{G}) \rightarrow \text{Mor}(\mathcal{G})$ is smooth, and the collection of composeable pairs $\text{Mor}(\mathcal{G}) \times_{\text{dom}, \text{cod}} \text{Mor}(\mathcal{G})$ is a smooth manifold and composition is smooth. More succinctly, a Lie groupoid is an *internal groupoid object* in the category Man of smooth manifolds ([nLa26]). An *étale groupoid* is a Lie groupoid where dom and cod are local diffeomorphisms.

Example 1.2.4. Let $M \times M \rightrightarrows M$ be the projections of a manifold M . Then this is a Lie groupoid with identity given by $x \mapsto (x, x)$. Another example is the first fundamental groupoid $\Pi_1(M)$ of a manifold M (recall morphisms are homotopy classes of paths).

1.2.1 The exponential map

Recall the flow of a vector field X is given by some $\gamma(t)$ satisfying

$$\frac{d\gamma(t)}{dt} = X(\gamma(t)), \quad \gamma(0) = p$$

for any $p \in M$. This locally has solutions on any manifold since it is an ODE, but in Lie groups it has solutions for all time on left-invariant vector fields since we can translate via left translation and the resultant equations still satisfy the ODE. This leads to a notion of the *exponential*.

Definition 1.2.5. Let G be a Lie group, and let $X \in T_e G$ be a left-invariant vector field. Then define $\exp^t X$ to be the flow $\gamma(t)$ of X satisfying

$$\frac{d\gamma}{dt} = X(\gamma(t)), \quad \gamma(t) = e$$

for $e \in G$ the identity. We then write $\exp = \exp^1$, so

$$\exp X = \gamma(1).$$

Lecture 3: h

Thu 29 Jan 2026 11:00

Oh god we went to a lecture on shifted symplectic structures last class that i did NOT understand because i dont know AG. i was following until he started computing homology and i dont know what (co?)homology theory he was using but it WASNT THE USUAL ONE

Let M a Lie group. Then $\text{Aut}_{\text{Man}}(M)$ is a Lie group as well. Then $T_1 \text{Aut}(M) = \mathfrak{diffeo}(M)$ is the Lie algebra. The convention is that $\text{Aut}(M)$ acts on M on the right as follows. Define $\cdot : \text{Aut}(M) \times M \rightarrow M$ by, for $x \in M$ and $\rho \in \text{Aut}(M)$,

$$x \cdot \rho := \rho^{-1}(x).$$

Remark 1.2.6. Since Lie groups have canonical isomorphisms $T_g G \cong T_e G$ for all $g \in G$, a metric on a Lie group is the same as an inner product on $T_e G$, and simply forcing left-invariance.

Definition 1.2.7. The Levi-Civita connection is the unique torsion-free metric-preserving connection. In local coordinates,

$$\nabla_i \partial_j = \Gamma_{ij}^k \partial_k.$$

Recall the geodesic equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0.$$

Then if X, Y, Z vector fields,

$$g(\nabla_X Y, Z) = \partial_X g(Y, Z) + \partial_Y g(X, Z) - \partial_Z g(X, Y) + g([X, Y], Z) - g([Y, Z], X) + g([Z, X], Y).$$

Lecture 4: ojdffd

Tue 03 Feb 2026 10:56

We are discussing the exponential $\exp : \mathfrak{g} \rightarrow G$ which relates the Lie algebra $\mathfrak{g} = T_e G$ to the Lie group G . We define $\exp(X)$ by integrating the left invariant vector field X .

Consider the exponential on Riemannian manifolds. There is a local geodesic so take a geodesic through p with X tangent to the geodesic. Then $\exp(X)$ is the point on the geodesic at time 1.

Ok for some reason $g([Y, Z], X) = g(\text{ad}_Z Y, X)$? A killing form is a bi-invariant form that need not be nondegenerate or positive-definite.

Definition 1.2.8. Let $\mathfrak{g}$ a Lie algebra. Then $\mathfrak{h} \subseteq \mathfrak{g}$ is a *Lie subalgebra* if it is a linear subspace closed under multiplication. Equivalently, $\mathfrak{h}$ is a Lie algebra and $\mathfrak{h} \hookrightarrow \mathfrak{g}$ is a Lie algebra homomorphism.

Definition 1.2.9. A Lie subgroup is a subgroup which is the image of an injective immersion into G .

Lecture 5: bro wtf is this class LMAO

Thu 05 Feb 2026 10:58

There is a correspondence between Lie subgroups and subalgebras given by $T_1 G$ and $\exp$. This correspondence also induces correspondences between normal subgroups and Lie ideals, Lie group homomorphisms and Lie algebra homomorphisms (recall $T_1 f = df = f_*$). Also recall a Lie algebra homomorphism is an $\mathbb{R}$ -linear map that commutes with the Lie bracket.

Theorem 1.2.10 (Closed subgroup). *Let $H \subseteq G$ be a closed subgroup. Then H is a submanifold.*

Theorem 1.2.11. *If a group homomorphism between Lie groups is continuous, then it is smooth.*

Also the diagram commutes

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\text{Ad}_g} & \mathfrak{g} \\ \exp \downarrow & & \downarrow \exp \\ G & \xrightarrow{\text{Conj}_g} & G \end{array}$$

for all $g \in G$. Here i dont know what Ad is but it induces $\text{ad}_X = [X, -]$ in some way.

Lecture 6: ???

Tue 10 Feb 2026 11:06

I learned something. The conjugation action $C_g : h \mapsto ghg^{-1}$ induces Ad_g via $\text{Ad}_g = \text{d}C_g$. Then this determines $G \rightarrow \text{End}(\mathfrak{g})$ via $g \mapsto \text{Ad}_g$, and the differential of this is ad_X .

$$\text{de}^X = \sum_{n \geq 0} \sum_{k=0}^{n-1} \frac{X^k \cdot \text{d}X \cdot X^{n-k-1}}{n!}.$$

I think here X is a matrix. Then this is

$$\int_0^1 e^{(1-t)X} \text{d}X e^{tX} \text{d}t.$$

or something.

Lecture 7: Cohomology

Thu 19 Feb 2026 12:07

Consider the *left-invariant* differential forms, meaning forms that act only on left-invariant vector fields $\Lambda^\bullet \mathfrak{g}$. Then the de Rham complex this yields is called the Cheveley-Eilenberg complex. In general, given vector fields $\{X_i\}_{1 \leq i \leq k}$ and a k -form ω , we have

$$\text{d}\omega(X_1, \dots, X_k) = \sum_i (-1)^i X_i(\omega(X_1, \dots, \hat{X}_i, \dots, X_k)) + \sum_{i < j} (-1)^{i+j} \omega([X_i, X_j], X_1, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_k).$$

However when only considering left-invariant vector fields on a Lie group, the first sum vanishes, so the differential is only the second term. If G is compact, then this agrees with the de Rham cohomology (over a field of characteristic 0).

Lecture 8: More cohomology

Thu 26 Feb 2026 11:00

Bibliography

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- [Huy04] Daniel Huybrechts. *Complex Geometry*. Springer UTX, 2004.
- [Lee00] John Lee. *Smooth Manifolds*. Springer, 2000.
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